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Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

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DEPARTMENT OF
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Bengaluru – 560059

Real-Time AI Governance Audit Pipeline

By

Project Report

Submitted by

NAME OF THE CANDIDATE(S)	USN
Aditya Ranjan	**1RV23AI008**
Arindam Gupta	**1RV23AI021**
Gnanendra Naidu N	**1RV23AI035**

in partial fulfillment for the requirement of 7th Semester

Stream Processing and Analytics (AI372IA)

Under the Guidance of

Dr. S. Anupama Kumar
Associate Professor
Department of AIML
RV College of Engineering
Bengaluru–59

Dr. Vijayalakshmi M.N
Associate Professor
Department of AIML
RV College of Engineering
Bengaluru–59

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CERTIFICATE

Certified that the project work titled **'Real-Time AI Governance Audit Pipeline'** is carried out by **Aditya Ranjan (1RV23AI008)**, **Arindam Gupta (1RV23AI021)**, and **Gnanendra Naidu N (1RV23AI035)**, who are bonafide students of RV College of Engineering, Bengaluru, in partial fulfillment of the curriculum requirement of 7th Semester **Stream Processing and Analytics** Experiential Learning during the academic year **2026–2027**. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the report. The report has been approved as it satisfies the academic requirements in respect of experiential learning work prescribed by the institution.

Signature of Faculty In-charge	**HoD**
Dept. of AIML, RVCE	**Dept. of AIML, RVCE**

External Examination

Name of Examiners	**Signature with date**
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ABSTRACT

This report presents the design, implementation, deployment evidence, and evaluation of a real-time AI governance audit pipeline. The system receives events from five producer domains, normalizes them into a versioned contract, evaluates sixteen governance rules with Kafka Streams, scores findings, persists evidence in PostgreSQL, and exposes reporting views through Tableau. Optional Bedrock-based Investigator and Evidence Reviewer agents support evidence-grounded analysis with bounded revision and human escalation. The report combines repository analysis with dated local and AWS evidence, including MSK, RDS, EKS, CloudWatch, Kafka-topic, and Tableau observations. Results include a labelled 72-event demonstration with 48 expected findings detected, measured recovery scenarios, and preserved audit traces. The implementation is presented as a reference architecture and does not claim legal conformity or general model accuracy.

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Chapter 1

INTRODUCTION

This chapter introduces the project context, describes the repository-based system under study, and explains how the report is organized.

1.1 Project Description

This report analyses the Git repository *AI-Governance-Audit-Pipeline*, a runnable reference implementation for collecting AI-governance evidence, detecting compliance conditions, scoring risk, persisting findings, and supporting evidence-grounded investigation and review. It follows the supplied SPA report format: formal chapters, numbered sections, source references, and an evidence appendix.

1.2 Report Organization

The analysis covers the source tree, schemas, rule catalogue, Kafka Streams processor, Python services, database schema, deployment files, evaluation material, and the dated evidence package. It does not claim that framework mappings establish legal conformity; the repository itself correctly treats them as evidence-routing decisions.

1.3 System overview

The system is a modular event-to-framework pipeline. Five independent adapters normalize source events into a canonical contract and publish to dedicated topics. A stateful Kafka Streams processor evaluates 16 rules, then a risk scorer and idempotent PostgreSQL sink complete the path. Tableau views expose operational and compliance reporting. Optional Amazon Bedrock Investigator and Evidence Reviewer agents add bounded, durable, advisory analysis. The governance framing follows contemporary risk-management guidance from NIST, ISO, the OECD, UNESCO, and the EU AI Act [2, 3, 4, 5, 6, 7].

Chapter 2

PROBLEM DEFINITION

This chapter defines the governance and engineering problem addressed by the pipeline and establishes the background needed to understand its relevance.

2.1 Problem Statement

AI systems generate evidence across identity, infrastructure, pipelines, model registries, and LLM gateways. Without a shared event contract and durable correlation, governance teams cannot reliably connect raw activity to controls, risk, evidence, and review outcomes. The project addresses this by combining source-specific adapters, event-time rules, risk scoring, persistence, and human-review boundaries. This emphasis on traceability and lifecycle controls is consistent with recent AI governance guidance [2, 4, 7, 9].

2.2 Literature Review

Recent literature frames AI governance as a lifecycle discipline rather than a final compliance checklist. NIST AI RMF 1.0 defines a voluntary risk-management structure around governing, mapping, measuring, and managing; its Generative AI Profile extends that structure to risks that are specific to generative systems [2, 3]. ISO/IEC 42001 formalizes the management-system perspective, while the OECD and UNESCO recommendations emphasize accountability, transparency, human oversight, and social impact [4, 5, 6]. The EU AI Act adds a risk-tiered regulatory context, but a framework mapping in this project is treated as routing metadata rather than a legal-conformity claim [7].

Security and systems literature identifies complementary operational concerns. OWASP and MITRE ATLAS catalogue prompt injection, data leakage, excessive agency, and adversarial techniques for AI-enabled applications [9, 10]; Google SAIF and OpenSSF emphasize secure development and supply-chain controls [11, 16]. Research on retrieval-augmented generation and autonomous agents highlights grounding, evaluation, citation quality, and bounded execution as open engineering challenges [19, 20]. Kafka Streams and Kubernetes documentation provide the implementation foundations for event-time state, joins, recovery, health checks, scaling, and disruption handling [12, 14].

The literature therefore provides principles, security taxonomies, and platform

mechanisms, but it does not by itself provide a single evidence path from heterogeneous producer activity to a stored finding, risk score, agent review, and human escalation. This project addresses that integration gap through a versioned event contract, stateful stream processing, durable PostgreSQL evidence, deterministic and comparative risk scoring, and bounded advisory agents.

Author (Year)	Journal / Conference	Method and Data	Key Contribution	Research Gap
Tabassi (2023) [2]	NIST AI RMF 1.0	Risk-management framework and lifecycle guidance	Govern–map–measure–manage structure for trustworthy AI	Does not prescribe an executable streaming evidence implementation
NIST (2024) [3]	NIST AI 600-1	Generative-AI risk profile and action-oriented guidance	Identifies GenAI risks and recommended controls	Requires system-specific telemetry, evidence, and review workflows
ISO/IEC (2023) [4]	ISO/IEC 42001 standard	AI management-system requirements	Establishes organizational ownership and continual improvement	Does not specify Kafka-based real-time detection or agent evidence storage
OECD (2024) [5]	OECD AI Principles	International policy principles and updates	Accountability, transparency, robustness, and human-centred values	Principles need operational control mappings and measurable evidence
European Parliament (2024) [7]	European Union regulation	Risk-tiered legal obligations	Defines governance and transparency expectations for high-risk AI	Regulation does not provide an implementation architecture for heterogeneous events
OWASP (2025) [9]	OWASP LLM Top 10	Threat taxonomy for LLM applications	Organizes prompt, data, agency, and supply-chain risks	Does not connect security findings to durable governance cases and review states
Gao et al. (2024) [19]	Survey literature / arXiv	Review of retrieval-augmented generation methods	Highlights grounding and evaluation requirements	Does not solve end-to-end governance evidence provenance
Wang et al. (2024) [20]	Survey literature / arXiv	Review of autonomous-agent architectures	Summarizes planning, tools, memory, and evaluation issues	Bounded revision and mandatory human escalation remain implementation concerns

Table 2.1: Literature review comparing governance, security, and agentic-AI sources with the project gap.

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Chapter 3

OBJECTIVES

This chapter states the primary objectives used to guide implementation and evaluate whether the project outcomes were achieved.

3.1 Primary Objectives

1. **Canonical event normalization:** Normalize five heterogeneous producer systems into one versioned governance event while retaining source identity, timestamps, actors, objects, and evidence metadata.
   Expected outcome: every accepted event follows a common contract and can be correlated across the pipeline.

2. **Executable governance detection:** Detect 16 governance controls using stateless predicates, event-time windows, and stateful joins.
   Expected outcome: configured violations produce deterministic, keyed findings with traceable rule identifiers.

3. **Persistent evidence and review identity:** Preserve event, finding, score, evidence, job, report, and review identity from ingestion through storage.
   Expected outcome: each finding can be reconstructed from durable records rather than transient logs.

4. **Explainable risk scoring:** Provide deterministic severity-frequency risk signals alongside an Isolation Forest comparison without replacing the rule baseline.
   Expected outcome: reviewers can distinguish the policy result from the comparative ML signal.

5. **Resilient and bounded operations:** Demonstrate replay, recovery, testing, traceability, and advisory agent review with human escalation when evidence remains incomplete.
   Expected outcome: documented workloads recover predictably and agent outputs remain reviewable.

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Chapter 4

TOOLS AND TECHNIQUES USED

This chapter identifies the software, infrastructure, and analytical techniques used in the project, with emphasis on the Kafka-based processing approach.

4.1 Tools

The project uses Python for adapters, scoring, evaluation, and agent workers; Java and Apache Kafka Streams for stateful stream processing; PostgreSQL for durable evidence; Docker Compose and Redpanda for local execution; Amazon MSK, RDS, EKS, CloudWatch, and Tableau for cloud deployment and reporting; Terraform and Kubernetes manifests for infrastructure automation; and XeLaTeX for this report. These tools correspond to the repository's service boundaries and deployment artefacts.

4.2 Techniques

The implementation applies the following techniques to keep findings traceable, reproducible, and operationally recoverable:

- schema-first event normalization and source-specific validation;
- event-time windows with grace periods for late records;
- stateful table-table and stream-table joins;
- deterministic rule-based detection and severity-frequency scoring;
- Isolation Forest comparison for an additional unsupervised signal;
- idempotent persistence, keyed findings, and independent consumer groups;
- evidence retrieval, structured agent validation, bounded revision, and human escalation.

Table 4.1 summarizes how the selected techniques are used in the implementation.

Technique	Application	Purpose
Schema-first processing	Versioned governance-event contract and mapper validation	Preserve provenance and prevent incompatible source payloads
Event-time processing	Five-minute windows with one-minute grace	Detect temporal conditions despite late or out-of-order events
Stateful joins	Approval/evaluation tables and promotion stream-table join	Combine related evidence before creating a finding
Risk scoring	Severity-frequency baseline plus Isolation Forest	Provide explainable baseline and comparative anomaly signal
Durable review	PostgreSQL upserts, evidence snapshots, and bounded agents	Make findings reproducible and review outcomes auditable

Table 4.1: Techniques used in the implemented governance pipeline.

4.2.1 Project pipeline and flow

Figure 4.1 summarizes the implemented path from source systems to governance reporting. Each producer adapter exposes an HTTP event endpoint, validates and maps source-specific input into the versioned governance-event contract, and publishes to its dedicated Kafka topic. Kafka Streams consumes the five topics, evaluates stateless predicates, applies event-time windows and joins, and emits keyed compliance findings. The risk scorer adds a deterministic baseline and an online Isolation Forest comparison. The PostgreSQL sink persists events, findings, evidence, and scores, while Tableau and the optional agents read durable outputs.

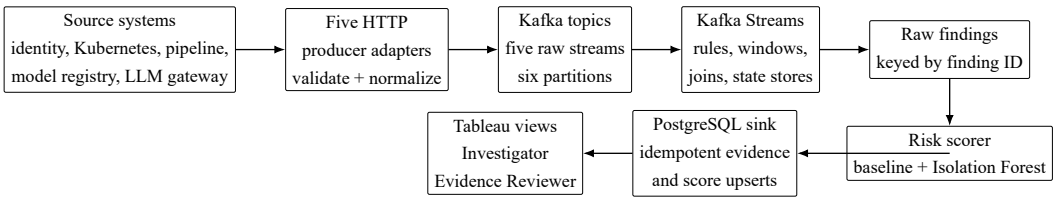


Figure 4.1: Implemented project pipeline from source events through Kafka Streams, scoring, persistence, and reporting.

4.2.2 Kafka features incorporated

Kafka feature	How it is incorporated	Project benefit
Dedicated topics	One raw topic per producer domain	Isolates source contracts and supports independent evolution
Partitioning and keys	Six partitions locally; actor, model, and finding keys	Enables parallel processing and deterministic correlation
Event-time processing	Timestamp extractor, five-minute window, one-minute grace	Handles late and out-of-order events correctly
State stores and joins	Materialized approval/evaluation state; table-table and stream-table joins	Combines related evidence before promotion findings
Consumer groups	Separate groups for rules, scoring, persistence, and evaluation	Permits independent replay and scaling
Durability and recovery	MSK replication factor three, minimum ISR two, committed offsets	Supports restart recovery and reconstructable audit trails
Typed serialization	JSON serializers/deserializers for events and findings	Preserves a readable, versioned message contract

Table 4.2: Kafka features incorporated in the project pipeline.

The project uses Kafka as an integration and recovery boundary rather than as a direct service-to-service call graph. The five producer topics isolate source contracts and allow each adapter to evolve independently. Keys are selected for the required state operation: actor IDs for failed-login windows and model keys for approval, evaluation, and promotion joins. The topology uses an event timestamp extractor so late or out-of-order records are evaluated against event time. A five-minute window with a one-minute grace period detects the fifth failed login without treating arrival time as the business event time.

Kafka Streams state stores materialize the model-approval state, model-evaluation state, combined model-governance evidence, and the failed-login window. The topology uses a table-table outer join to combine approval and evaluation evidence, then a stream-table left join to check a production promotion against that evidence. State-

less rule findings and stateful findings are merged into one raw-finding topic, and the finding ID becomes the output key for downstream correlation. JSON serdes keep the event and finding records typed while retaining a human-readable contract.

The deployment separates consumer groups for rule processing, risk scoring, persistence, and evaluation. This allows each downstream function to replay from its own committed offsets and prevents a reporting slowdown from blocking detection. The local Compose configuration creates six partitions per topic, runs two Kafka Streams threads, and uses Redpanda as the local Kafka-compatible broker. The AWS configuration raises durability through a three-broker MSK cluster, replication factor three, and minimum in-sync replica policy. These features support parallelism, replay, fault recovery, and independent scaling while keeping the audit trail reconstructable.

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Chapter 5

IMPLEMENTATION

This chapter explains how the repository implements the architecture, contracts, rules, scoring, persistence, agent integration, deployment, and operational controls.

5.1 High-level architecture

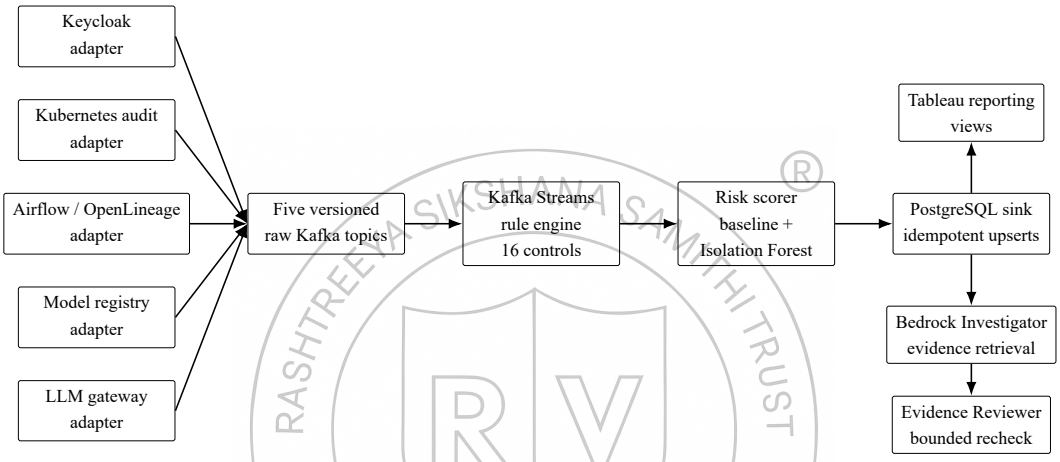


Figure 5.1: High-level architecture derived from the repository services, deployment manifests, and documented event flow.

The architecture separates producers, stream processing, scoring, storage, and reporting. Local Compose uses Redpanda and PostgreSQL; AWS deployment definitions provide MSK, EKS, RDS, ECR, IAM, VPC, and security groups. Kafka topics and JSON contracts are the primary integration boundary. The design follows documented event-stream processing and cloud-native operational patterns [12, 13, 14, 15].

5.2 Module responsibilities

Component	Responsibility
Producer adapters	Five standalone HTTP services for Keycloak, Kubernetes audit, Airflow/OpenLineage, model registry, and LLM gateway.
Stream processor	Java Kafka Streams topology with event-time windows, joins, rule catalog, and finding creation.

Risk scorer	Online Isolation Forest comparison with a deterministic severity-frequency baseline.
PostgreSQL sink	Idempotent event and finding upserts plus indexed evidence and reporting views.
Agents	Bedrock Investigator and Evidence Reviewer with durable jobs, structured contracts, and bounded revision.
Deployment	Docker Compose, Kubernetes manifests, Terraform infrastructure, and Tableau setup.

5.3 Data flow

```
source -> adapter -> raw topic -> canonical event -> rule engine
       -> finding -> risk scorer -> PostgreSQL -> Tableau / agents
```

5.4 Data Sources and Contracts

5.5 Producer groups

Each producer has its own adapter, health/readiness/metrics endpoints, image, deployment, scaling policy, IAM role, and topic. This reduces coupling and permits source-specific validation while maintaining a common downstream contract.

5.6 Canonical event contract

The versioned governance-event schema captures identity, source, timestamps, actor, object, action, attributes, and evidence metadata. Mappers remove source-specific variation while preserving useful provenance. Safe dead-letter handling prevents malformed input from silently entering the main path. This is aligned with current software supply-chain and observability practice [16, 17].

5.7 Topics and integration

The repository defines one raw topic per producer and separate consumer groups for rule processing, scoring, persistence, and evaluation. Six partitions and independent consumer groups support parallelism; offsets, keys, and event IDs enable reconciliation.

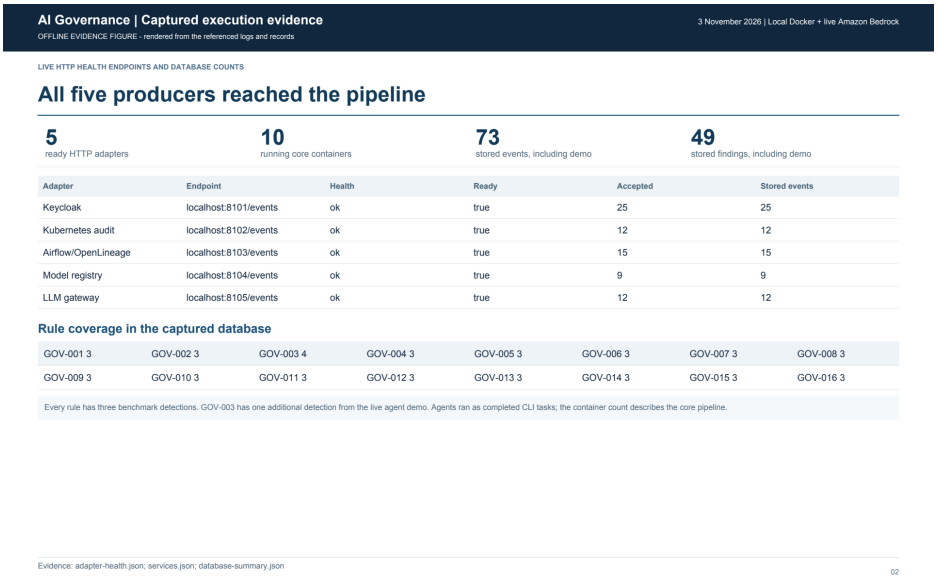


Figure 5.2: Captured health and accepted-event evidence for the five producer adapters.

5.8 Rule Engine and Framework Mapping

5.9 Executable controls

The 16-rule catalogue in `config/rules.yml` maps each control to NIST AI RMF 1.0, ISO/IEC 42001:2023, EU AI Act, NIST SP 800-53 Rev. 5, and GDPR. The Java implementation contains stateless predicates, a five-minute failed-login window with one-minute grace, and stateful promotion joins. The separation between a control mapping and a legal conclusion follows the risk-management approach in [2, 4, 7].

5.10 Finding record

Findings retain rule ID, severity, triggering event identity, timestamps, framework labels, evidence, and processing metadata. This makes a control result inspectable rather than a transient alert.

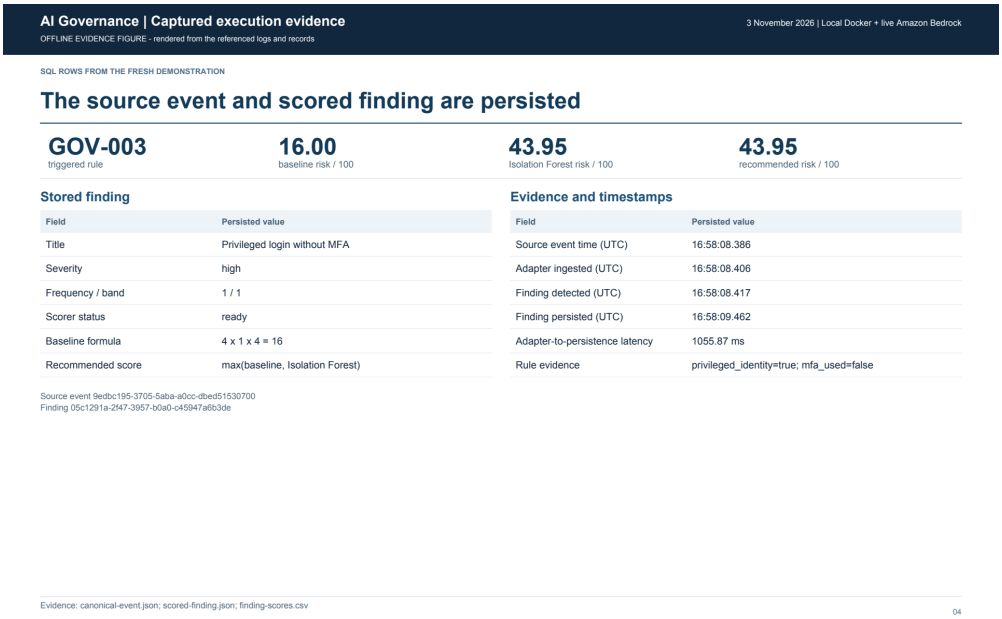


Figure 5.3: Persisted finding evidence showing the triggering event, rule, scores, and timestamps.

5.11 Risk Scoring, Storage and Reporting

5.12 Risk scoring

The deterministic baseline combines a 1–5 frequency bucket with severity and produces a normalized 0–100 score. An online Isolation Forest maintains bounded history and stores its score, delta, readiness status, and recommended score beside the baseline. The ML signal is comparative and does not remove the explainability of the rule result. The implementation uses the established scikit-learn API and keeps the baseline visible for auditability [18].

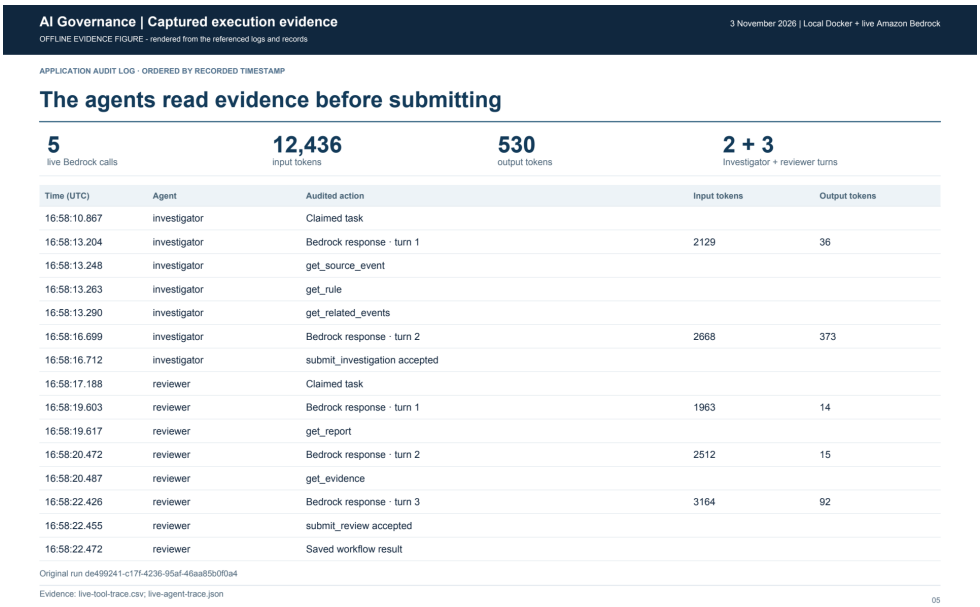
5.13 Persistence model

PostgreSQL stores raw events, findings, evidence, risk scores, jobs, agent reports, reviews, and audit history. Upserts and indexes support idempotency and case retrieval. Tableau views cover framework status, risk trend, incident history, and pipeline latency.

5.14 Agent integration

The Investigator retrieves eligible evidence and writes a structured report. The Evidence Reviewer checks citations and completeness, can request at most one revision, and may terminate in human review. Reports and evidence snapshots remain

immutable; agents do not remediate systems. This bounded pattern reflects current agent observability and responsible-agent guidance [21, 22, 20].



AI Governance | Captured execution evidence

OFFLINE EVIDENCE FIGURE - rendered from the referenced logs and records

3 November 2026 | Local Docker + live Amazon Bedrock

APPLICATION AUDIT LOG · ORDERED BY RECORDED TIMESTAMP

The agents read evidence before submitting

5	12,436	530	2 + 3
live Bedrock calls	input tokens	output tokens	Investigator + reviewer turns

Time (UTC)	Agent	Audited action	Input tokens	Output tokens
16:58:10.867	investigator	Claimed task		
16:58:13.204	investigator	Bedrock response · turn 1	2129	36
16:58:13.248	investigator	get_source_event		
16:58:13.263	investigator	get_rule		
16:58:13.290	investigator	get_related_events		
16:58:16.699	investigator	Bedrock response · turn 2	2668	373
16:58:16.712	investigator	submit_investigation accepted		
16:58:17.188	reviewer	Claimed task		
16:58:19.603	reviewer	Bedrock response · turn 1	1963	14
16:58:19.617	reviewer	get_report		
16:58:20.472	reviewer	Bedrock response · turn 2	2512	15
16:58:20.487	reviewer	get_evidence		
16:58:22.426	reviewer	Bedrock response · turn 3	3164	92
16:58:22.455	reviewer	submit_review accepted		
16:58:22.472	reviewer	Saved workflow result		

Original run de499241-c17f-4236-95af-46aa85b0f0a4

Evidence: live-tool-trace.csv; live-agent-trace.json

05

Figure 5.4: Live tool trace for the Investigator and Evidence Reviewer.

5.15 Deployment and Operations

5.16 Local operation

The documented local path uses Docker Desktop with at least 6 GB available: copy `.env.example`, start Compose, inject labelled evaluation events, and export a report. Redpanda stands in for MSK and a PostgreSQL container stands in for RDS.

5.17 AWS operation

Terraform provisions a VPC, three-broker MSK, EKS, RDS, ECR, security groups, and workload IAM. Kubernetes manifests define services, probes, HPAs, PDBs, and consumer scaling. The deployment instructions correctly require image publication, secret creation, placeholder replacement, and an evaluation before recording results.

5.18 Resilience and controls

Idempotent writes, retrying producers, replication factor three, minimum ISR two, consumer groups, and reconciliation checks address loss and restart scenarios. Operational scripts pause and resume services; failure procedures distinguish the local single-broker demonstration from AWS broker failover.

Chapter 6

RESULTS AND DISCUSSION

This chapter presents the observed functional, resilience, cloud, dashboard, and agent-evaluation evidence, followed by interpretation and limitations.

6.1 Verification coverage

The repository includes Python, Java, mapper, integration, settings, reviewer, topology, scorer, and evaluation tests. Requirements traceability links each major requirement to implementation and verification. The 3 November evidence package records 83 agent tests passed in 9.65 seconds.

6.2 Measured functional run

Scenario	Events	Exp./Obs.	Precision	Recall	Throughput
Baseline	72	48/48	1.0000	1.0000	130.90/s
Processor replacement	72	48/48	1.0000	1.0000	11.50/s
Broker restart	72	48/48	1.0000	1.0000	11.48/s
Independent adapters	72	48/48	1.0000	1.0000	667.12/s

Table 6.1: Repository-reported local evaluation results.

The fresh 3 November evidence also reports 72 synthetic source-shaped events, 48 expected findings across all 16 rules, p50 353.84 ms, p95 1,878.88 ms, p99 2,181.69 ms, and 39.04 events/s over the observed persistence interval. These are small-workload measurements, not general model-accuracy claims. The project dashboard in Figure 6.1 presents the same reporting concepts in a compact view. The governance pipeline combines 16 deterministic rules, an Isolation Forest scorer, and two cooperating Bedrock agents: the Investigator retrieves evidence and drafts a report, while the Evidence Reviewer checks citations and requests a bounded correction or human review. The scorer begins training after 32 findings, retrains after each additional 25 findings, retains up to 1,000 recent findings, and recommends the higher of the baseline and ML scores when the model is ready.

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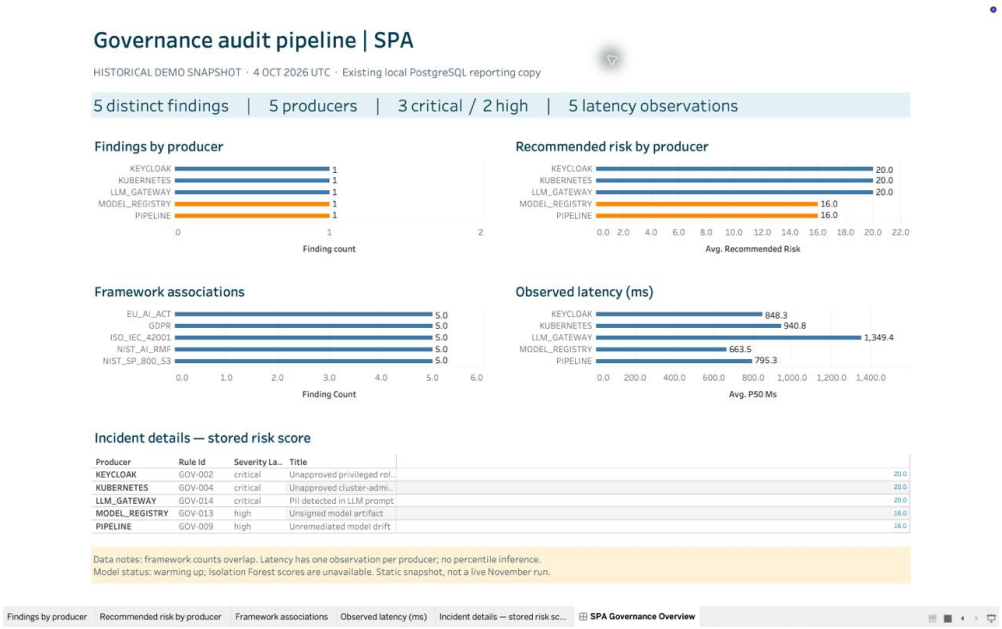


Figure 6.1: Supplied Tableau-style governance dashboard reference showing findings, risk, framework associations, latency, and incident details.

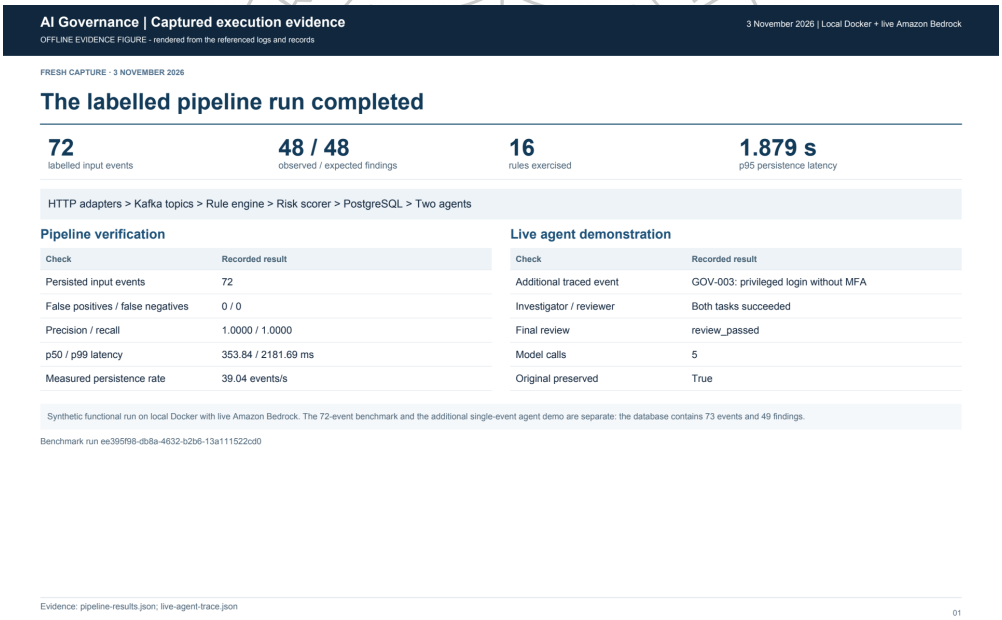


Figure 6.2: Pipeline benchmark evidence for the labelled functional run.

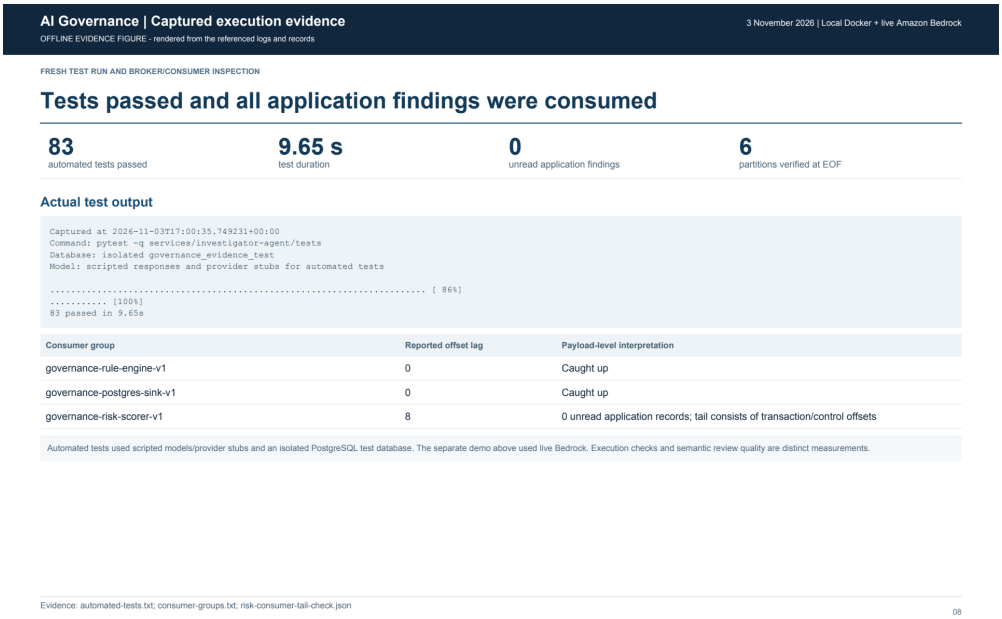


Figure 6.3: Automated tests and consumer-tail verification.

6.3 Agentic AI Evaluation and Limitations

6.4 Observed outcomes

One fresh individual case completed investigation and review with `review_passed`; the original report and evidence snapshot were preserved. A saved correction example demonstrates cited feedback, one revision, recheck, and terminal `needs_human_review`. CloudWatch evidence for the selected window records five Bedrock invocations, 12,436 input tokens, 530 output tokens, and 1,476.4 ms average latency.

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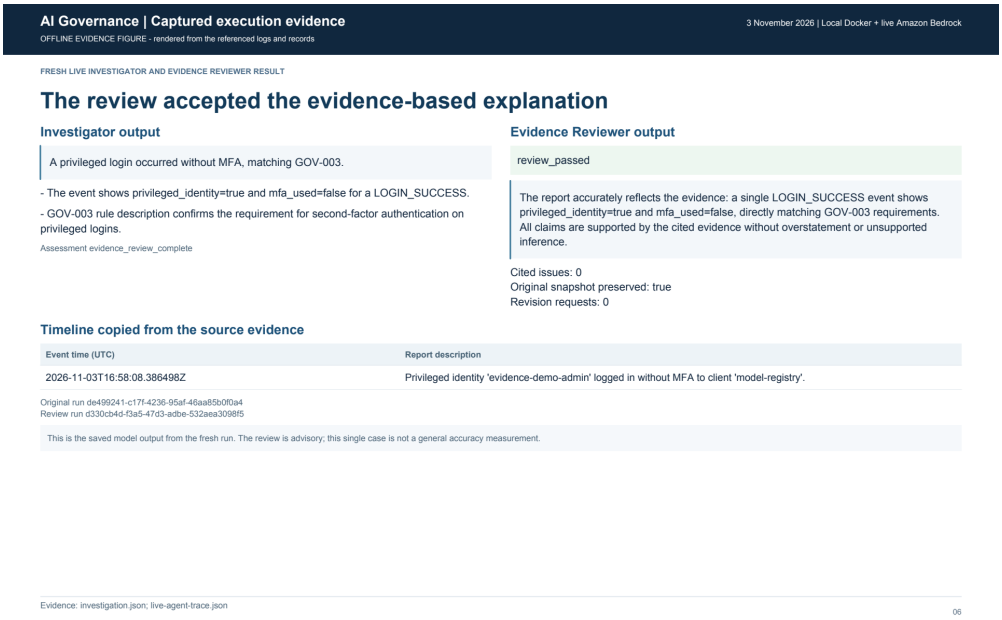


Figure 6.4: Fresh reviewed report with preserved evidence.

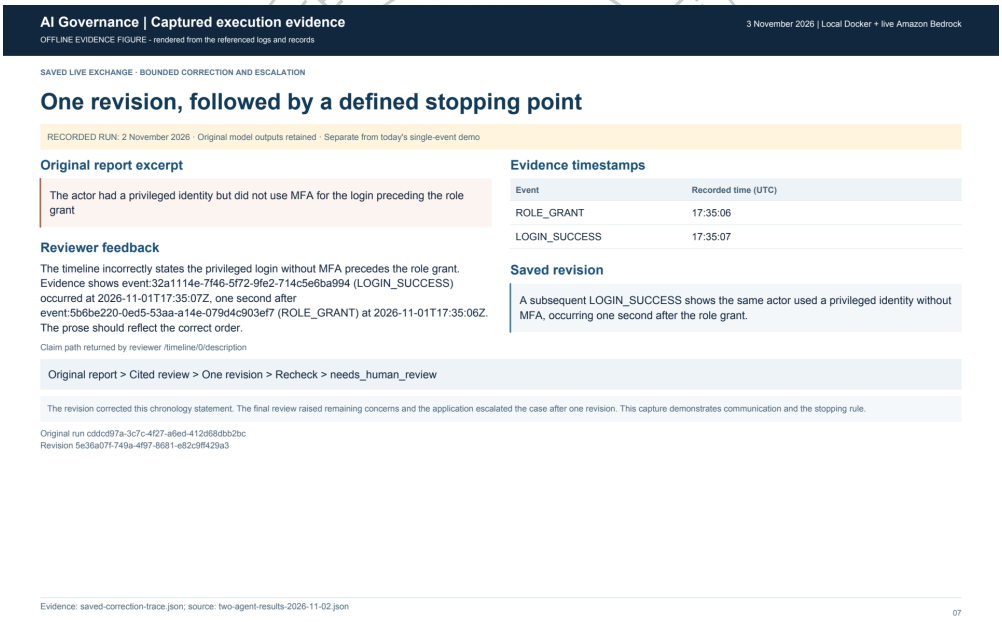


Figure 6.5: Saved correction, revision, and human-escalation example.

6.5 Limitations

The reviewer is advisory and must not be treated as an automatic approval gate. The evidence package contains selected demonstrations rather than corpus-wide semantic-quality evaluation. Framework labels are routing metadata, not legal conclusions. The AWS metrics are account/model/region aggregates for the selected window, while the application trace provides case-level correlation.

6.6 Discussion

The measured results indicate that the implemented objectives were met for the documented workloads: the labelled pipeline recovered all expected findings in the baseline and fault-injection demonstrations, while the evidence package preserves the separate agent outcomes and their limitations. The AWS screenshots corroborate that the cloud resources were provisioned and observable, but they do not replace a controlled production readiness review. The EKS control-plane logs also show stale metrics API discovery messages, which should be treated as an operational follow-up rather than silently classified as a successful health result.

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Chapter 7

CONCLUSION

This chapter summarizes the engineering contribution, the evidence-supported outcomes, and the practical significance of the completed pipeline.

The project is a coherent, runnable governance-audit platform with unusually strong traceability for a reference implementation. Its strongest engineering decisions are independent adapters, explicit contracts, event-time/stateful processing, idempotent persistence, measured recovery tests, immutable evidence, and bounded agent workflows. The main adoption requirement is operational discipline: execute the documented evaluation in the target environment, retain evidence, and preserve human review for uncertain or corrected agent outcomes.

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Chapter 8

FUTURE ENHANCEMENTS

This chapter identifies extensions that would improve production readiness, evaluation depth, operational resilience, and governance coverage.

Future work should add a production-scale semantic-quality benchmark for the agents, formal schema compatibility checks across release versions, stronger alerting for stale Kubernetes metrics APIs, automated evidence retention policies, multi-region disaster recovery, continuous Tableau refresh validation, and a governed approval workflow for changes to framework mappings. A larger labelled corpus would also allow calibration of risk thresholds beyond the current demonstration workloads.

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APPENDIX: CLOUD AND REPORTING EVIDENCE

The following screenshots were supplied with the project analysis package and are included as operational evidence. Figures 8.1–8.8 document cloud infrastructure and service observability. Figures 8.9–8.10 document the Tableau reporting layer. The screenshots are descriptive evidence of the captured environment and should be read with the repository's dated provenance files.

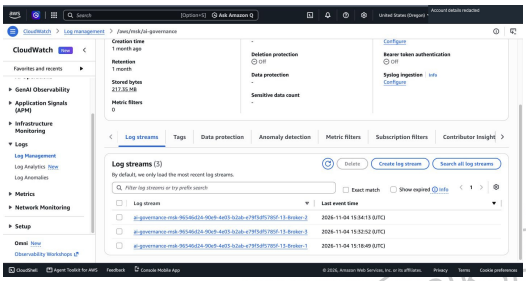


Figure 8.1: CloudWatch log group and MSK broker log streams.

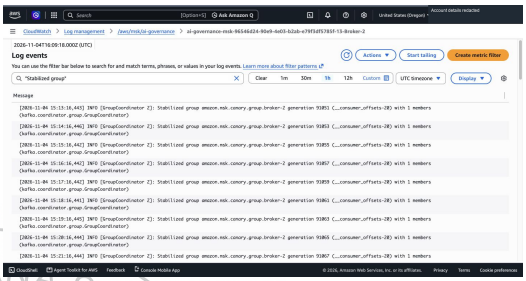


Figure 8.2: Kafka broker logs showing stabilized consumer coordination.

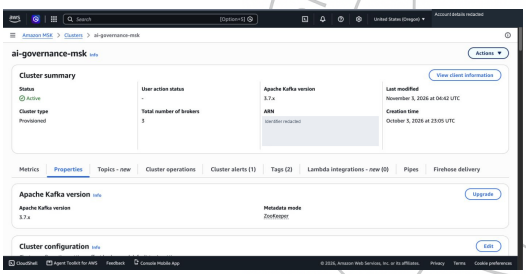


Figure 8.3: Active provisioned three-broker MSK cluster.

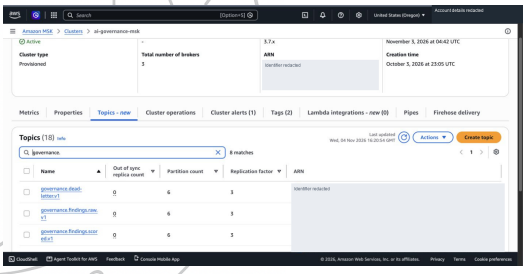


Figure 8.4: Governance topics with six partitions and replication factor three.

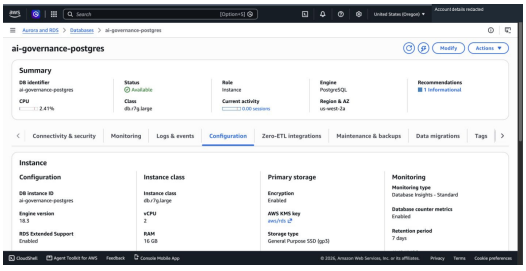


Figure 8.5: RDS PostgreSQL configuration for durable evidence.

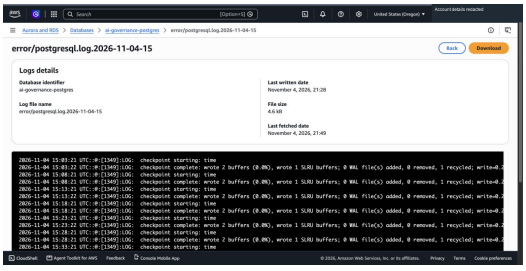


Figure 8.6: PostgreSQL checkpoint activity in service logs.

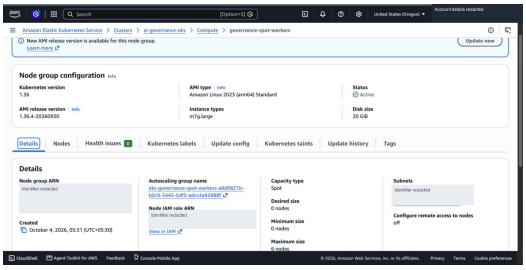


Figure 8.7: EKS worker-node configuration.

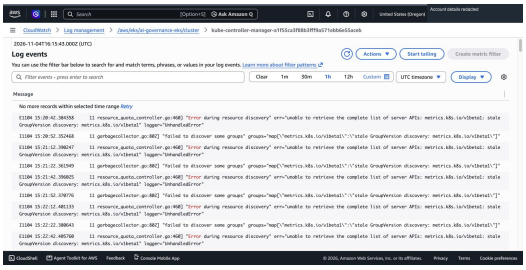


Figure 8.8: EKS control-plane metrics API messages requiring follow-up.

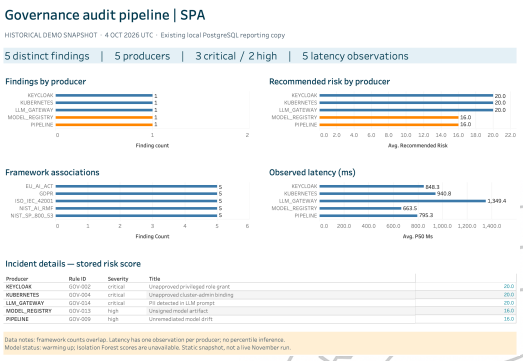


Figure 8.9: Tableau governance dashboard.

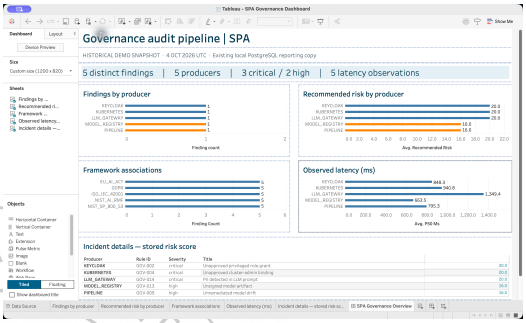


Figure 8.10: Tableau Desktop authoring view.

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